

C.W

04.10.2026

Graph

Def: Graph $G = \{V, E\}$

vertices
vertexes
Node

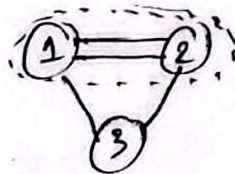
Edges

Definition: არ არის არა

Multigraph:

• multiple edges.

• Ex:



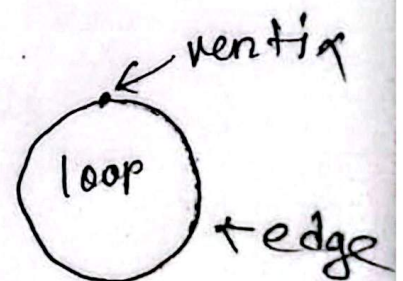
• ყველა vertex-ზე არის multiple edges

თმე,

pseudograph:

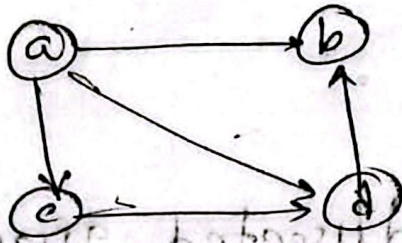
• Single vertex

• Minimum one loop



- Node - 1 ~~node~~ edges connected
2nd - 2nd - 1st degree.

Ex:



$$\deg(a) = 3$$

$$\deg(b) = 2$$

$$\deg(c) = 2$$

$$\deg(d) = 3$$

in degree:

$$\deg^-(a) = 0$$

$$\deg^-(b) = 2$$

$$\deg^-(c) = 1$$

$$\deg^-(d) = 2$$

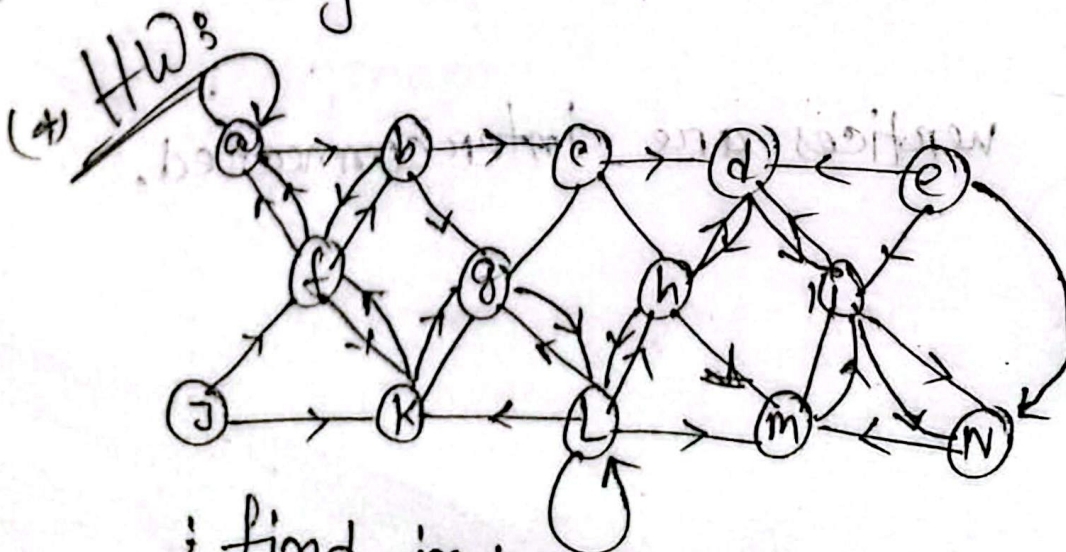
out degree:

$$\deg^+(a) = 3$$

$$\deg^+(b) = 0$$

$$\deg^+(c) = 1$$

$$\deg^+(d) = 2$$



i find in deg & out deg

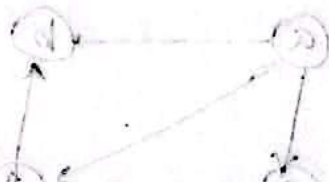
Directed graph:

- There will be directions for edges.



Directed multigraph:

- Combination of directed graph and multigraph.



Weighted graph:

- Every graph has a weight associated with it.
- edge has a value or weight.

CC Graph:

- all vertices are interconnected.

